ECE 3340

Numerical Methods

Exam 3A

Name:____

Random Numbers _________ **/ 20**

Derivatives _________ **/ 20**

Differential Equations _________ **/ 30**

Numerical Integration _________ **/ 30**

Total _________ / 100

Random Numbers. Calculate the parameters for an LCG that will provide the maximum period for a 3-bit random number generator. Show each value in the sequence starting with $x_0 = 1$. **[20 points]**

$\boldsymbol{m} =$ 8

$\boldsymbol{c} =$ 1, 3, 5, 7

$\boldsymbol{a} =$ 5

Derivatives. Calculate the gradient of the explicit function $\boldsymbol{\nabla\phi}(2, -1)$ represented by the sample points in the grid using 2nd order finite differences. The gradient of a function is defined as it's partial derivative along each axis: **[20 points]**

	-4	-3	-2	-1	0	1	2	3	4
-3	3.2	2.1	1.8	0.8	1.0	0.8	1.8	2.1	3.2
-2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2
-1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
0	2.0	1.0	0.0	-1.0	-2.0	-1.0	0.0	1.0	2.0
1	2.1	1.4	0.4	-0.7	-1.0	-0.7	0.4	1.4	2.1
2	2.2	1.8	0.7	0.4	0.0	0.4	0.7	1.8	2.2
3	3.2	2.1	1.8	0.8	1.0	0.8	1.8	2.1	3.2

$$\boldsymbol{\nabla\phi} = \begin{bmatrix} d\phi/dx \\ d\phi/dy \end{bmatrix}$$

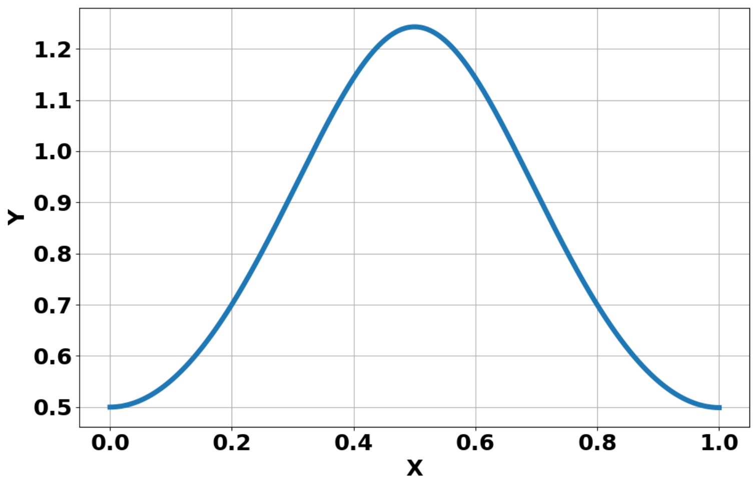
$$\frac{d\phi}{dx} \approx \frac{1.4 - (-0.7)}{2(1)} = \frac{2.1}{2} = 1.05$$

$$\frac{d\phi}{dy} \approx \frac{0.0 - (0.7)}{2(1)} = \frac{-0.7}{2} = -.35$$

Explicit Integration. Solve the following differential equation using explicit integration

$$f(x,y) = \frac{dy}{dx} = \sin(2\pi x)\, e^{y}$$

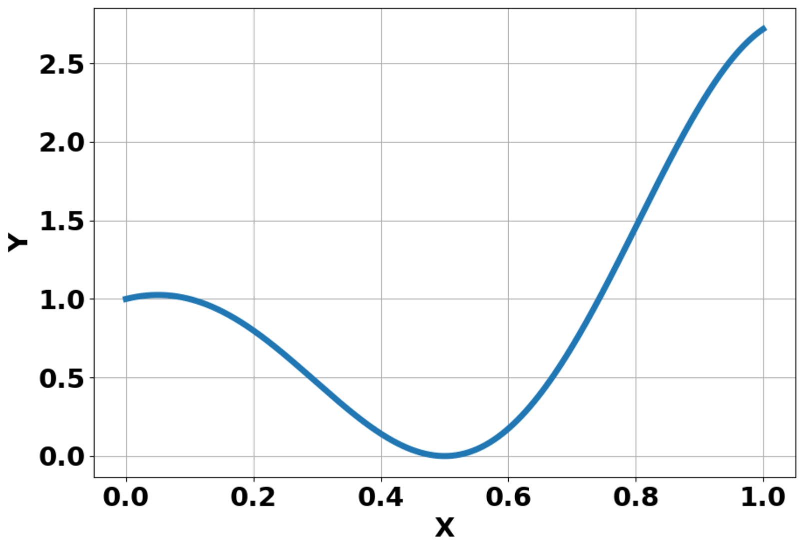
given the initial condition $(x,y) = (0, 0.5)$ and $\Delta x = 0.2$. Plot the resulting function alongside the real result below:



Use Euler's Method (4 steps) **[40 points]**

x	y	$f(x,y)$	E_r
0	0.5	0	
0.2	0.5 + 0.2(0)=0.5	1.57	
0.4	0.5 + 0.2(1.57)=.814	1.33	
0.6	0.5 + 0.2(1.33)=1.08	-1.73	
0.8	0.5 + 0.2(-1.73)=.876		

Numerical Integration. Approximate the following integral using Simpson's rule and the Trapezoid rule. Use the same three points for the Trapezoid rule (using 2 trapezoids). **Show your work. [40 points]**



$$\int_0^1 \cos^2(\pi x)\, e^x dx$$

$$S = \frac{1}{6}\left[f(0) + \frac{1}{4}f(0.5) + f(1)\right] = \frac{1}{6}[1 + 0 + 2.72] = \frac{3.72}{6} = 0.62$$

$$T = \frac{1}{2}\left[\frac{1}{2}[f(0) + f(0.5)] + \frac{1}{2}[f(0.5) + f(1)]\right] = \frac{1}{4}[1 + 0 + 0 + 2.72]$$

$$= \frac{3.72}{4} = 0.93$$